

HOMEWORK 7 FOR MATH 2860 (ELEMENTARY DIFFERENTIAL EQUATIONS)

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Due at 10:00 EST in class

Question Q1. The complex-valued function $f(t)$ depends on real variable t :

$$f(t) = \exp [(-1 + i2\pi)t]$$

- (a) Compute the real and imaginary part of the function $\mathbf{re}f(t)$ and $\mathbf{im}f(t)$.
- (b) Sketch $\mathbf{re}f(t)$ and $\mathbf{im}f(t)$ for $0 \leq t < 3$ to accurately represent growth/decay of each function at $t = 0$, and points of intersection with the horizontal axis.
- (c) For what values t is the function $f(t)$ purely real (imaginary part is 0) and for what values is it purely imaginary (real part is 0)?

Show all your work.

Question Q2. An unforced response of an oscillator can be modeled using the ODE

$$m\ddot{x}(t) + c\dot{x}(t) + kx(t) = 0,$$

where m is the mass, c is the damping coefficient, and k the spring stiffness.

- (i) For $m = 4$ and $k = 9$ and undetermined $c \geq 0$, find the roots of the characteristic equation.
- (ii) Find the ranges of values of c such that the general solution is:
 - (i) decaying and shows no oscillation,
 - (ii) decaying and has oscillations.
- (iii) Choose one value for each range you determined and write out the formula for the general solution.

Show all your work.